

Random variables and distribution functions

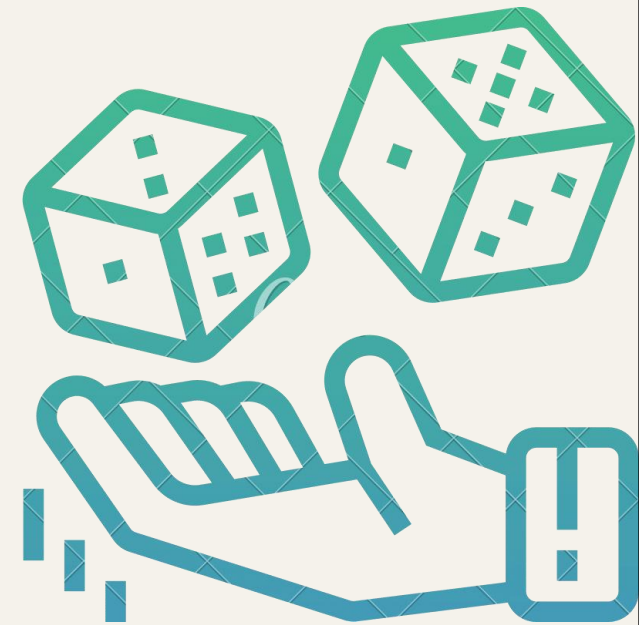
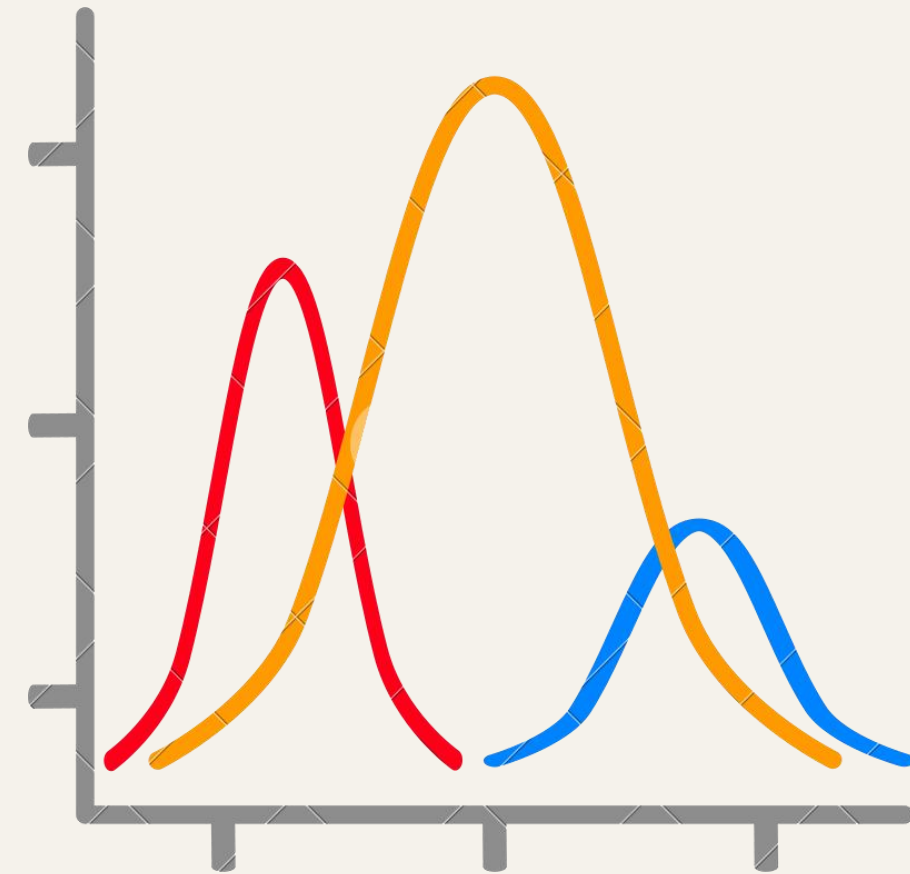


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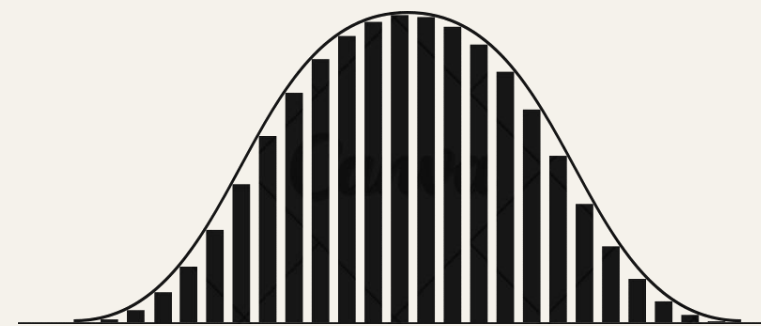
$\times_n$

$\int \times$

$P(A | B)$

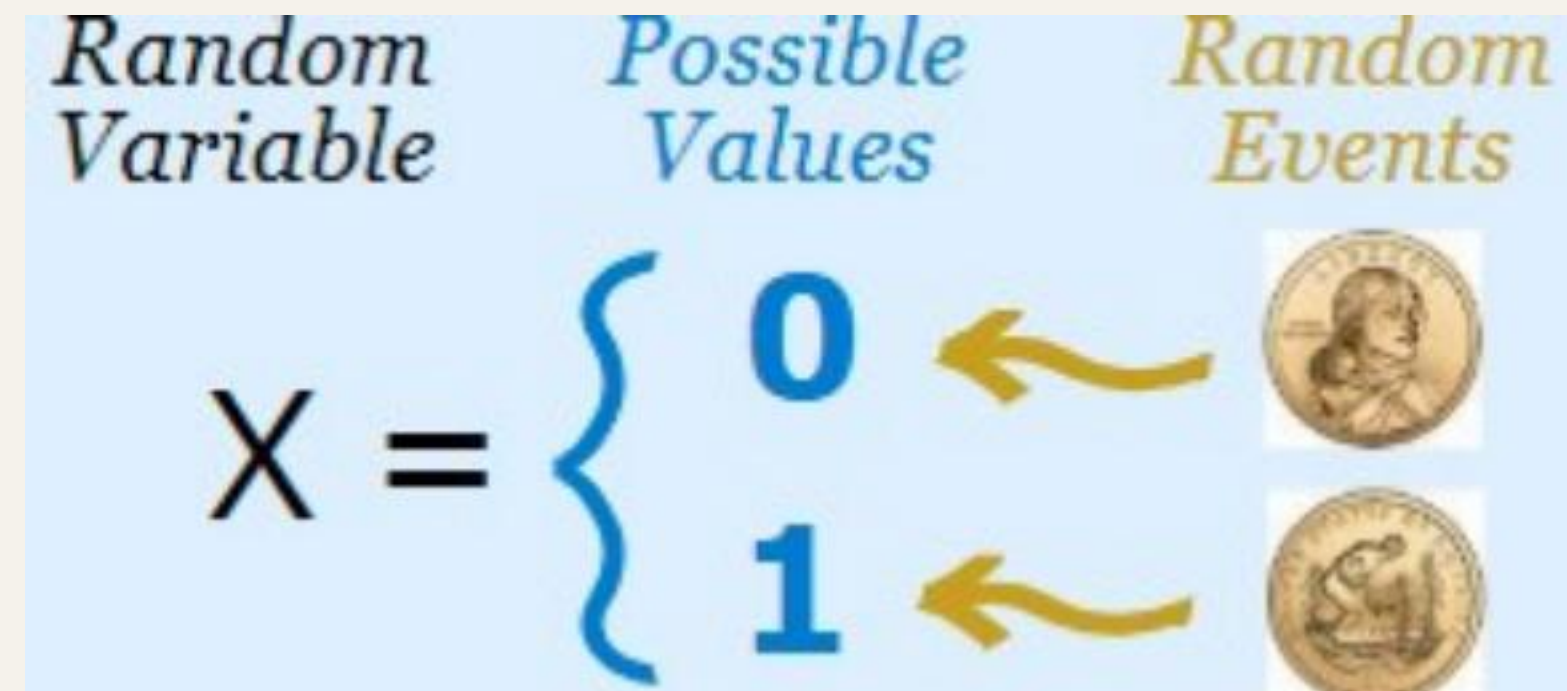


$f(x)$



RANDOM VARIABLES

- A random variable is a set of possible values of a random experiment
- A random variable is usually represented using capital letters X



RANDOM VARIABLE

● Discrete Random Variable

A random variable is said to be discrete if it is allowed to take only limited number of values which can be listed.

Rolling an unbiased die, the possible values that the random variable X can take are from the list $\{1,2,3,4,5,6\}$

● Continuous Random Variable

A random variable is said to be continuous if it is allowed to assume any value within a specified range.

Height of person in cms., speed of vehicle in km/hr

Probability Distribution of a Random Variable

- Let X be a discrete random variable of a finite sample space X .
- Let $x_1, x_2, \dots, x_n$ be the values of random variable X with probabilities $p_1, p_2, \dots, p_n$

Then the function $P(X = x)$ is called probability distribution or probability mass function (p.m.f.) of random variable X if

i) $0 \leq P(x_i) \leq 1$ Where, $i = 1, 2, \dots, n$

ii) $\sum_{i=1}^n P(x_i) = 1$



Random Experiment

- A random experiment is a process by which we observe something uncertain. After the experiment, the result of the random experiment is known.
- An outcome is a result of a random experiment.
- The set of all possible outcomes is called the sample space.

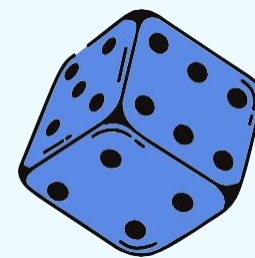
Example

1. Tossing a coin

Sample Space $S = \{\text{heads}, \text{tails}\}$ or $\{H, T\}$

2. Rolling a die

Sample Space $S = \{1, 2, 3, 4, 5, 6\}$



Example 1:

Suppose three fair coins are tossed simultaneously and random variable X denote the number of heads then the possible values of X are 0,1,2,3

$S = \{ HHH, TTT, HTT, THT, TTH, THH, HTH, HHT \}$

Now,

$$P(X=0) = \frac{1}{8}, P(X=1) = \frac{3}{8}, P(X=2) = \frac{3}{8}, P(X=3) = \frac{1}{8}$$

Also

$$\sum_{i=0}^3 P(x_i) = 1$$

Hence,

$X = x$	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

is a probability distribution of X

Example 2:

The probability distribution function of a discrete function of a discrete random variable X is given by;

X	-2	-1	0	1	2
$P(X = x)$	0.1	0.15	0.2	0.15	0.4

Find :

i. $P(X \leq 0)$ *ii.* $P(X \geq -1)$

Also obtain the probability distribution $y = x^2$

Solution:

Given:

$$\begin{aligned} i. \quad P(X \leq 0) &= P(X = -2, -1, 0) \\ &= P(X = -2) + P(X = -1) + P(X = 0) \\ &= 0.1 + 0.15 + 0.2 \end{aligned}$$

$$\therefore P(X \leq 0) = 0.45$$

ii. $P(X \geq -1) = P(X = -1, 0, 1, 2)$
 $= P(X = -1) + P(X = 0) + P(X = 1) + P(X = 2)$
 $= 0.15 + 0.2 + 0.15 + 0.4$

$\therefore P(X \geq -1) = 0.9$

Now,
 $y = x^2$

x	-2	0	1	2
y	4	0	1	4

$\therefore$ Values of y are 0, 1 and 4

Now;
 $P(Y = 0) = P(X = 0) = 0.2$
 $P(Y = 1) = P(X = -1) + P(X = 1) = 0.15 + 0.15 = 0.3$
 $P(Y = 4) = P(X = -2) + P(X = 2) = 0.1 + 0.4 = 0.5$
 $\therefore$ The probability distribution of $y = x^2$ is;

y	0	1	4
$P(Y = y)$	0.2	0.3	0.5

Example 3:

Find K in each of the following so that $P(X = x)$ is a probability mass function(p.m.f) of X .

$X = x$	-1	0	1	2	3
$P(X = x)$	K	$2K$	$3K$	$4K$	$5K$

Solution:

Since $P(X = x)$ is a probability mass function(p.m.f) of X then

$$\sum_{x=-1}^3 P(x) = 1$$

$$\therefore K + 2K + 3K + 4K + 5K = 1$$

$$\therefore 15K = 1$$

$$\therefore K = \frac{1}{15}$$

Example 4:

The p.m.f of a random variable X is zero except at the points $X=0,1,2$. At these point $P(0) = 3C^3, P(1) = 4C - 10C^2, P(2) = 5C - 1$

i. Determine C

ii. Find $P(X < 1), P(1 < X \leq 2), P(0 < X \leq 2)$

Solution:

Since $\sum_x P(x_i) = 1$

$$\sum_{x=0}^2 P(x) = 1$$

$$\therefore P(0) + P(1) + P(2) = 1$$

$$\therefore 3C^3 + 4C - 10C^2 + 5C - 1 = 1$$

$$\therefore 3C^3 - 10C^2 + 9C - 2 = 0$$

$$\therefore (C - 1)(3C^2 - 7C + 2) = 0$$

$$\therefore (C - 1)(C - 2)(3C - 1) = 0$$

$$\therefore C = 1, 2, \frac{1}{3}$$

Here, C is the probability

$\therefore C = 1$ and $C = 2$ is not possible

$$\therefore C = \frac{1}{3}$$

$\therefore$ The Probability distribution of X is

x	0	1	2
$P(X = x)$	1/9	2/9	2/3

$$\text{ii. } P(< 1) = P(X = 0) = \frac{1}{9}$$

$$P(1 < X \leq 2) = P(X = 2) = \frac{2}{3}$$

$$\begin{aligned} P(0 < X \leq 2) &= P(X = 1, 2) \\ &= P(X = 1) + P(X = 2) = \frac{1}{9} + \frac{2}{3} \end{aligned}$$

$$\therefore \mathbf{P(0 < X \leq 2) = \frac{8}{9}}$$

Questions For Practice:

1. Verify whether the following function can be regarded as p.m.f of the random variable X .

$X = x$	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.3	0.2	0.2

2. Find value k if probability mass function of X is given as $P(X = x) = k(x + 1)^2, x = 0,1,2,3$

3. Let X be the discrete random variable with following probability distribution.

X	0	1	2	3	4
$P(X=x)$	0.1	0.2	0.3	0.15	0.25

Find

- i) $P(X < 1)$ ii) $P(X \geq 3)$ iii) $P(1 < X < 4)$ iv) $P(2 \leq X \leq 3)$ v) $P(X > 1 | X \leq 3)$

Solution

Q.1

1. $X=x$	1	2	3	4	5
$P(X=x)$	0.1	0.2	0.3	0.2	0.2

check whether given f^n is pmf

i) $P(X=x) \geq 0$

ii) $\sum_{x \in X} P(X=x) = 1$

$\therefore P(X) \geq 0$ for each value of x

$$\sum P(X=x) = 1$$

$$0.1 + 0.2 + 0.3 + 0.2 + 0.2 = 1$$

$$\therefore \sum P(X=x) = 1$$

$\therefore$ Given function is a pmf

Solution

Q.2

$$2. \quad P(X=x) = k(x+1)^2 \quad x=0,1,2,3$$

$$\therefore \sum_{x \in X} P(X=x) = 1$$

$$\sum_{x=0}^3 k(x+1)^2 = 1$$

$$k[(0+1)^2 + (1+1)^2 + (2+1)^2 + (3+1)^2] = 1$$

$$k[1 + 2^2 + 3^2 + 4^2] = 1$$

$$k[1 + 4 + 9 + 16] = 1$$

$$k[30] = 1$$

$$k = \frac{1}{30}$$

$$\therefore P(X=x) = \frac{1}{30}(x+1)^2 \quad x=0,1,2,3$$

$$= 0$$

O.W

Solution

Q.3

$$\text{i) } P(X < 1) = P(X = 0) \\ = 0.1$$

$$\text{ii) } P(X \geq 3) = P(X = 3) + P(X = 4) \\ = 0.15 + 0.25 \\ = 0.40$$

$$\text{iii) } P(1 < X < 4) = P(X = 2) + P(X = 3) \\ = 0.3 + 0.15 \\ = 0.45$$

$$\text{iv) } P(2 \leq X \leq 3) = P(X = 2) + P(X = 3) \\ = 0.3 + 0.15 \\ = 0.45$$

$$\text{v) } P(X > 1 | X \leq 3) = \frac{P(X > 1 \cap X \leq 3)}{P(X \leq 3)} \\ = \frac{P(X = 2) + P(X = 3)}{1 - P(X > 3)}$$

$$= \frac{0.3 + 0.15}{1 - 0.15}$$

$$= \frac{0.45}{0.85}$$

$$= 0.5294$$

Cumulative Distribution Function

Let X be a discrete random variable with values $x_1, x_2, \dots, x_n$ and corresponding probabilities $p_1, p_2, \dots, p_n$ then cumulative distribution function of X is given as $F(X = x) = P(X \leq x_i)$
Where $i = 1, 2, \dots, n$

Properties of c.d.f of X :

- a) $0 \leq F(x) \leq 1, \forall x \in X$
- b) $F(-\infty) = 0$ and $F(\infty) = 1$
- c) If $a \leq b$ then $F(a) \leq F(b)$
- d) $P(a < x \leq b) = F(b) - F(a)$

Questions for Practice:

1. Consider the following p.m.f. of X

$X = x$	1	3	5
$P(X = x)$	$3/5$	$3/10$	$1/10$

Obtain c.d.f of X.

2. Following is the c.d.f of random variable X.

$X = x$	1	2	3	4	5	6
$F(X = x)$	0.2	0.37	0.48	0.62	0.85	1

- i) Find p.m.f of X
- ii) $P(2 < X < 5)$
- iii) $P(X \leq 5/X > 3)$
- iv) Find P.m.F of $Y=3X+1$

3. From a collection of 10 bulbs containing 3 defectives, a sample of 4 bulbs are drawn at random. Let the random variable X denote the number of defective bulbs. Find the probability distribution of X. Also find $P(X \leq 1)$ and $P(0 < X < 2)$.

Probabiility Density Function

- The equation used to describe a continuous probability distribution is called a probability density function.
- Since the continuous random variable is defined over a range of values, the graph of the density function will also be continuous over the range.
- The area bounded by the curve of the density function and the x-axis is equal to 1, when computed over the domain of the variable.
- The probability that a random variable assumes a value between a and b is equal to the area under the density function bounded by a and b.

Definition:

Let X be a continuous random variable. The function $f(x)$ is called the probability density function of X if it satisfies the following:

i) $f(x) \geq 0, \forall x \in R$

ii) $\int_{-\infty}^{\infty} f(x). dx = 1$

Note: If X takes values in $a < x < b$ then $\int_a^b f(x). dx = 1$

Example 1:

A continuous random variable X has a probability density function(pdf);

$$f(x) = 3x^2, \quad 0 \leq x \leq 1$$

Find “a” and “b” such that

i. $P[X \leq a] = P[X > a]$

ii. $P[X > b] = 0.05$

Solution:

$$f(x) = 3x^2, \quad 0 \leq x \leq 1 \text{ ----- Given}$$

i. Since ; $P[X \leq a] = P[X > a]$

$$\therefore \int_0^a f(x)dx = \int_a^1 f(x)dx$$

$$\int_0^a 3x^2 dx = \int_a^1 3x^2 dx$$

$$3\left[\frac{x^3}{3}\right]_0^a = 3\left[\frac{x^3}{3}\right]_a^1$$

$$a^3 = 1 - a^3$$

$$2a^3 = 1$$

$$\therefore a^3 = \frac{1}{2}$$

$$\therefore \mathbf{a} = \left(\frac{\mathbf{1}}{\mathbf{2}}\right)^{1/3}$$

$$\text{ii. } P[X > b] = 0.05$$

$$\therefore \int_b^1 f(x)dx = 0.05$$

$$\int_b^1 3x^2 dx = 0.05$$

$$3\left[\frac{x^3}{3}\right]_b^1 = 0.05$$

$$1 - b^3 = 0.05$$

$$\therefore b^3 = 1 - 0.05 = 0.95$$

$$\therefore b^3 = \frac{95}{100} = \frac{19}{20}$$

$$\therefore \mathbf{b} = \left(\frac{\mathbf{19}}{\mathbf{20}}\right)^{1/3}$$

Cumulative Distribution Function for Continuous Random Variable

If X is a continuous random variable then the Cumulative Distribution Function $F(x)$ is defined as

$$F(X) = P[X \leq x] = \int_{-\infty}^x f(x) \quad \text{for} \quad -\infty < x < \infty$$

Example 1:

$$\text{If } f(x) = xe^{-\frac{x^2}{2}}, \quad x \geq 0$$
$$= 0 \quad x < 0$$

Find its cumulative distribution function

Solution:

$$F(X=x) = P(X \leq x)$$
$$= \int_0^x f(x) dx$$
$$= \int_0^x x e^{-\frac{x^2}{2}} dx$$

$$\text{Let } \frac{x^2}{2} = t \quad \frac{2x dx}{2} = dt$$
$$\Rightarrow x dx = dt$$

Range

for	x	t
	0	0
	x	$x^2/2$

$$F(X=x) = \int_0^{x^2/2} e^{-t} dt$$
$$= \left. \frac{e^{-t}}{-1} \right|_0^{x^2/2}$$
$$= -e^{-t} \Big|_0^{x^2/2}$$
$$= -e^{-\frac{x^2}{2}} - (-e^{-0})$$
$$= -e^{-\frac{x^2}{2}} + 1$$
$$F(X=x) = 1 - e^{-\frac{x^2}{2}} \quad x \geq 0$$
$$= 0 \quad x < 0$$

Questions for Practice:

1. The probability density function of continuous random variable is given by

$$f(x) = \frac{x+2}{18}, -2 < x < 4$$
$$= 0, \text{ otherwise}$$

Find (i) $P(X < 1)$ (ii) $P(-1 < X < 1)$

2. Find k , if the function f defined by $f(x) = kx$, $0 < x < 2$ is the p.d.f of random variable X .

3. A random variable X has the p.d.f. given by

$$f(x) = kx^2(1-x)^2, 0 < x < 1$$
$$= 0, \text{ otherwise}$$

Find value k . Also find $P(X < \frac{1}{2})$

Solution

Q.1

$$\begin{aligned} \text{i) } P(X < 1) &= \int_{-2}^1 f(x) dx \\ &= \int_{-2}^1 \frac{x+2}{18} dx \\ &= \frac{1}{18} \left[\frac{x^2}{2} + 2x \right]_{-2}^1 \\ &= \frac{1}{18} \left[\left(\frac{1}{2} + 2 \right) - \left(\frac{4}{2} + 2(-2) \right) \right] \\ &= \frac{1}{18} \left[\frac{5}{2} - (2 - 4) \right] \\ &= \frac{1}{18} \left[\frac{5}{2} - (-2) \right] \\ &= \frac{1}{18} \left[\frac{5}{2} + 2 \right] \\ &= \frac{1}{18} \times \frac{20}{2} \\ &= \frac{5}{9} \end{aligned}$$

$$\begin{aligned} \text{ii) } P(-1 < X < 1) &= \int_{-1}^1 f(x) dx \\ &= \int_{-1}^1 \frac{x+2}{18} dx \\ &= \frac{1}{18} \left[\frac{x^2}{2} + 2x \right]_{-1}^1 \\ &= \frac{1}{18} \left\{ \left(\frac{1}{2} + 2 \right) - \left(\frac{1}{2} - 2 \right) \right\} \\ &= \frac{1}{18} \left(\frac{5}{2} - \left(-\frac{3}{2} \right) \right) \\ &= \frac{1}{18} \left(\frac{5}{2} + \frac{3}{2} \right) \\ &= \frac{1}{18} \times \frac{8}{2} \\ &= \frac{2}{9} \end{aligned}$$

Solution

Q.2

$$\int_0^2 f(x) = 1$$

$$\int_0^2 kx = 1$$

$$\left. \frac{kx^2}{2} \right|_0^2 = 1$$

$$k \left[\frac{2^2}{2} - \frac{0^2}{2} \right] = 1$$

$$k [2] = 1$$

$$k = \frac{1}{2}$$

$$f(x) = \frac{1}{2}x \quad 0 < x < 2$$

$$= 0 \quad \text{o.w.}$$

Joint Probability Law

If X and Y are two random variables then the probability distribution that defines the simultaneous occurring of both events is called as **Joint Probability Distribution**.

Important Points:

- A joint probability is a statistical measure used to calculate the likelihood of two events taking place at the same time.
- Joint probability is also called the intersection of two or more events.

For Discrete :

The joint probability mass function of two discrete random variables X,Y is:

$$P_{XY}(x,y)=P(X=x,Y=y)=P((X=x) \text{ and } (Y=y)).$$

where i. $0 \leq P_{XY}(x,y) \leq 1$

ii. $\sum_x \sum_y P(x,y) = 1$

Marginal PMF:

$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

Marginal probability refers to the probability of a single event regardless of the outcome of the other variable.

Consider two random variables X and Y with joint PMF given in Table

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$

Find

- a. Marginal PMF of X and Y
- b. $P(X=0, Y \leq 1)$
- c. Find $P(Y=1|X=0)$

Solution

	$Y=0$	$Y=1$	$Y=2$	Total
$X=0$	$1/6$	$1/4$	$1/8$	$13/24$
$X=1$	$1/8$	$1/6$	$1/6$	$11/24$
Total	$7/24$	$10/24$	$7/24$	1

i) Marginal Pmf of X

$$P_X(x) = \sum_{y \in Y} P(x, y)$$

$$= \sum_{y=0}^2 P(x, y)$$

X	0	1
$P_X(x)$	$13/24$	$11/24$

Marginal Pmf of Y

$$P_Y(y) = \sum_{x \in X} P(x, y)$$

$$= \sum_{x=0}^1 P(x, y)$$

Y	0	1	2
$P(Y=y)$	$7/24$	$10/24$	$7/24$

Solution

b- $P(X=0, Y \leq 1)$

$$\begin{aligned} P(X=0, Y \leq 1) &= P(X=0, Y=0) + P(X=0, Y=1) \\ &= \frac{1}{6} + \frac{1}{4} \\ &= \frac{10}{24} \end{aligned}$$

c) $P(Y=1 | X=0) = \frac{P(Y=1, X=0)}{P(X=0)}$

$$= \frac{1/4}{13/24}$$

$$= \frac{6}{13}$$

The joint pmf of random variabes X and Y is given by

$$P(x,y) = \frac{x+y}{21} \quad x = 1,2,3; y=1,2$$

$$0 \quad \text{otherwise}$$

Obtain i. Marginal Pmf of X

ii. Marginal Pmf of Y

iii. Examine whether X and Y are independent

Solution

a) Marginal Pmf of X

$$P_X(x) = \sum_{y \in Y} P(x, y)$$
$$= \sum_{y=1}^2 \frac{x+y}{21}$$

$$= \frac{x+1}{21} + \frac{x+2}{21}$$

$$P_X(x) = \frac{2x+3}{21} \quad x=1, 2, 3$$

X	1	2	3
$P(X=x)$	5/21	7/21	9/21

b. Marginal Pmf of Y

$$P_Y(y) = \sum_{x \in X} P(x, y)$$
$$= \sum_{x=1}^3 \frac{x+y}{21}$$

$$= \frac{1+y}{21} + \frac{2+y}{21} + \frac{3+y}{21}$$

$$P_Y(y) = \frac{6+3y}{21} \quad y=1, 2$$

Y	1	2
$P(Y=y)$	9/21	12/21

Solution

ii) Examine whether X and Y are independent

$$P(X=x, Y=y) = P(X=x) * P(Y=y)$$

$$\therefore \text{LHS} = P(X=x, Y=y)$$

$$= \frac{x+y}{21}$$

$$\text{RHS} = P(X=x) * P(Y=y)$$

$$= \left(\frac{2x+3}{21} \right) \left(\frac{6+3y}{21} \right)$$

$$= \frac{12x+6xy+18+9y}{21}$$

$$= \frac{12x+6xy+9y+18}{21}$$

$\therefore X$ and Y are not independent

Questions:

1. A fair coin is tossed 2 times. Let X denote the number of heads occurring in the last two tosses and Y denote the total number of heads occurring in the two tosses. Find the joint probability distribution of X and Y . Obtain conditional probability distribution of Y given $X=1$.
2. Suppose that X and Y have joint pdf
$$P(x,y) = \frac{xy^2}{39}$$
for the points $(1,2)$ $(1,3)$ $(2,2)$ $(2,3)$ and it is 0 otherwise.
Find the i. Marginal Probability Distribution of X and Y
ii. Are X and Y independent

Practice Questions

1. A two dimensional random variable (X,Y) have a bivariate distribution given by

$$P(X=x, Y=y) = \frac{x^2+y}{32} \quad ; \text{for } x=0,1,2,3 \text{ and } y = 0,1$$

- a. Find the Marginal distributions of X and Y
- b. Find $P(X=1)$, $P(Y = 0)$, $P(X=2, Y=1)$, $P(X=2|Y=1)$

An advisor looks over the schedules of his 50 students to see how many mathematics and science courses each has registered for in the coming semester. He summarizes his results in the table

		Number of Mathematics Courses(X)		
Number of Science Course(Y)		0	1	2
	0	11	6	4
	1	9	10	3
	2	5	0	2

Obtain

- i. Joint Probability Distribution of X and Y
- ii. Marginal Probability Distribution of X
- iii. Conditional Probability Distribuion of Y given $X = 2$
- iv. Are variables X and Y independent

Solution

Q	y \ x	X			Total
		0	1	2	
	0	11	6	4	21
	1	9	10	3	22
	2	5	0	2	7
	Total	25	16	9	Total=50

i) Joint Prob of x and y

y	X			Total = $P_y(y)$
	0	1	2	
0	11/50	6/50	4/50	21/50
1	9/50	10/50	3/50	22/50
2	5/50	0	2/50	7/50
$P_x(X) = \text{Total}$	25/50	16/50	9/50	1

ii) Marginal Distⁿ of X

$$P_x(x) = \sum_{y \in Y} P(x, y)$$

X	0	1	2
$P(X=x)$	25/50	16/50	9/50

Solution

iii) Conditional Prob of y given $x=2$

$$P(y|x=2)$$

$$\therefore P(y|x) = \frac{P(y \cap x)}{P(x)}$$

$$P(y|x=2) = \frac{P(y=y \cap x=2)}{P(x=2)}$$

$$P(y=0|x=2) = \frac{P(y=0 \cap x=2)}{P(x=2)}$$

$$= \frac{4/50}{9/50}$$

$$= \frac{4}{9}$$

$$P(y=1|x=2) = \frac{P(y=1 \cap x=2)}{P(x=2)}$$

$$= \frac{3/50}{9/50} = \frac{3}{9}$$

$$P(y=2|x=2) = \frac{P(y=2 \cap x=2)}{P(x=2)}$$

$$= \frac{2/50}{9/50}$$

$$= \frac{2}{9}$$

$y=y$	0	1	2
$P(y x=2)$	$\frac{4}{9}$	$\frac{3}{9}$	$\frac{2}{9}$

Solution

iv) Are x and y independent

$$P(X=x, Y=y) = P(X=x) * P(Y=y)$$

$\therefore$ For $X=0$ $Y=1$

$$\text{LHS} = P(X=0, Y=1) = 6/50$$

$$\begin{aligned} \text{RHS } P(X=0) &= 25/50 & P(Y=0) &= \\ P(Y=1) &= 22/50 \end{aligned}$$

$$\& P(X=0, Y=1) = P(X=0) P(Y=1)$$

$$\frac{6}{50} = \frac{25}{50} \times \frac{22}{50}$$

$$\frac{6}{50} \neq \frac{11}{50}$$

LHS $\neq$ RHS ; x and y are not independent

Joint Probability Density Function

If continuous random variables X and Y are defined on the same sample space S , then their **joint probability density function (joint pdf)** is a piecewise continuous function, denoted $f(x, y)$, that satisfies the following.

1. $f(x, y) \geq 0$, for all $(x, y) \in \mathbb{R}^2$

2. $\iint_{\mathbb{R}^2} f(x, y) dx dy = 1$

3. $P((X, Y) \in A) = \iint_A f(x, y) dx dy$, for any $A \subseteq \mathbb{R}^2$

Marginal Probability Density Function

Suppose that continuous random variables X and Y have joint density function $f(x,y)$. The marginal pdf's of X and Y are respectively given by the following:

Marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy, \quad \text{for all } x,$$
$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx, \quad \text{for all } y.$$

Practice Questions

Q.1

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} x + cy^2 & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

a. Find the constant c .

b. Find $P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2})$.

c. Marginal PDF of X's and Y's

Transformation of One Dimensional Random Variable

For Discrete Case

Let X be discrete random variable with pmf $P(x)$. If the transformation $Y=g(X)$ is defined for all values within the range of X . Then the pdf of $Y = g(X)$ is obtained by considering the 2 cases

Case 1:

When relation is one to one $Y=g(X)$ for any value of X i.e for one value of X we have one value of Y

Case 2:

When relation is not one-one $Y=g(X)$ for any value of X i.e. for one value of X we have more than one values of Y

Examples:

1. For the following PMF of X

$$P_x(X) = \frac{x^2}{10} \quad x = 1, 2, 3, 4, 5$$

Find

a. Pmf Of $Y = 3X + 1$

b. $P(Y)$

2. For the following PMF of X

$$P_x(X) = \begin{cases} \frac{1}{12} & x=2, 4, 6 \\ 0 & \text{o.w} \end{cases}$$

Find the pmf of Y where $Y = 2X$

Solution

$$P_X(x) = \frac{x^2}{10} \quad x=1,2,3,4,5$$

X	1	2	3	4	5
P(X=x)	1/10	4/10	9/10	16/10	25/10

i) pmf of $Y=3X+1$

$$\text{for } X=1 \quad Y=4$$

$$X=2 \quad Y=7$$

$$X=3 \quad Y=10$$

$$X=4 \quad Y=13$$

$$X=5 \quad Y=16$$

$$\therefore \text{for } X=1 \Rightarrow Y=4$$

$$P(X=1) = P(Y=4) = \frac{1}{10}$$

$$X=2 \Rightarrow Y=7$$

$$P(X=2) = P(Y=7) = \frac{4}{10}$$

$$X=3 \Rightarrow Y=10$$

$$P(X=3) = P(Y=10) = \frac{9}{10}$$

$$X=4 \Rightarrow Y=13$$

$$P(X=4) = P(Y=13) = \frac{16}{10}$$

$$X=5 \Rightarrow Y=16$$

$$P(X=5) = P(Y=16) = \frac{25}{10}$$

Y	4	7	10	13	16
P(Y=y)	1/10	4/10	9/10	16/10	25/10

Solution

$$2^{\circ} P_X(x) = \frac{1}{12} \quad x=2,4,6$$
$$= 0 \quad \text{o.o.w.}$$

x	2	4	6
$P(X=x)$	$1/12$	$1/12$	$1/12$

$$Y = 2X$$

$$Y = 2X$$

$$\text{For } X=2 \quad Y=4$$

$$X=4 \quad Y=8$$

$$X=6 \quad Y=12$$

$$\text{For } X=2 \quad Y=4 \Rightarrow P(X=2) = P(Y=4) = \frac{1}{12}$$

$$\text{For } X=4 \quad Y=8 \Rightarrow P(X=4) = P(Y=8) = \frac{1}{12}$$

$$\text{For } X=6 \quad Y=12 \Rightarrow P(X=6) = P(Y=12) = \frac{1}{12}$$

y	4	8	12
$P(Y=y)$	$1/12$	$1/12$	$1/12$

Question

Let X denote number of heads when coin is tossed twice. Find the prob distribution of $Y=(X-1)^2$

Let X denote number of heads when coin is tossed twice
 $S = \{HH, HT, TH, TT\}$
 $X = 0, 1, 2$

X	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

i) $Y = (X-1)^2$

$$\begin{array}{ll} X=0 & Y=1 \\ X=1 & Y=0 \\ X=2 & Y=1 \end{array}$$

$$\begin{array}{ll} \text{For } X=0 & Y=1 \\ X=2 & Y=1 \end{array}$$

Here, for two different values of X we get same values of Y

i.e. $X=0$ and $X=2$ we get $Y=1$

$$\begin{aligned} P(Y=1) &= P(X=0) + P(X=2) \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{2}{4} \end{aligned}$$

$$\text{For } X=1 \Rightarrow Y=0 \quad \therefore P(X=1) = P(Y=0) = \frac{2}{4}$$

3. Pmf of Y is

Y	0	1
$P(Y=y)$	$\frac{2}{4}$	$\frac{2}{4}$

SUMS FOR Practice

1. Let X denote number of tails when we toss a coin simultaneously. Find the probability distribution function of Y where

i. $Y = 3X - 1$

ii. $Y = (1 + X)^3$

2. Let X denote number of heads when coin is tossed thrice. Find the prob distribution of $Y = (X - 1)^2$

Transformation of Continuous Random Variable

Let X be a continuous random variable with pdf $f(x)$. If the transformation $y = v(x)$ is

- continuously differentiable
- either increasing or decreasing
- for all values within the range of X then the pdf of $Y=v(x)$ is given by
- $\mathbf{g(y) = f(x(y)) \left| \frac{dy}{dx} \right|}$

1. If X is a random variable with pdf
 $f(x) = 1 ; 0 < x < 1$
 0 o.w

Find the Pdf for Y if $Y = 2X$

2. If X is a random variable with pdf
 $f(x) = 2x ; 0 < x < 1$
 0 o.w

Find the Pdf for Y if $Y = 8X^3$

3. If X is a random variable with pdf
 $f(x) = x/12 ; 1 < x < 5$
 0 o.w

Find the Pdf for Y if $Y = 2X - 3$

Solution

$$10 \quad f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & 0 > w \end{cases}$$

Pdf of $y = 2x$

$$f(y) = f(x) \times \left| \frac{dx}{dy} \right|$$

$$\therefore y = 2x \Rightarrow x = \frac{y}{2}$$

For Range of $y = 2x$

x	y
0	0
1	2

$$\therefore \left| \frac{dx}{dy} \right| = \left| \frac{d}{dy} \left(\frac{y}{2} \right) \right|$$

$$= \frac{1}{2}$$

$$f(y) = f(x) \times \left| \frac{dx}{dy} \right|$$

$$= 1 \times \left| \frac{1}{2} \right|$$

$$f(y) = \frac{1}{2} \quad 0 < y < 2$$

$$= 0 \quad 0 > w$$

Solution

$$2^{\circ} f(x) = 2x \quad 0 < x < 1 \\ = 0 \quad 0 = w$$

pdf of $y = 8x^3$.

$$f(y) = f(x) \times \left| \frac{dx}{dy} \right|$$

$$\because y = 8x^3 \Rightarrow x = \sqrt[3]{\frac{y}{8}} = \left(\frac{y}{8}\right)^{\frac{1}{3}}$$

$$\frac{dx}{dy} = \frac{1}{3} \left(\frac{y}{8}\right)^{\frac{1}{3}-1} \times \frac{1}{8}$$

$$\frac{dx}{dy} = \frac{1}{24} \left(\frac{y}{8}\right)^{\frac{2}{3}}$$

Range for y

x	y
0	0
1	8

$$f(y) = f(x) \times \left| \frac{dx}{dy} \right|$$

$$= 2x \times \frac{1}{24} \left(\frac{y}{8}\right)^{\frac{2}{3}}$$

$$\because x = \left(\frac{y}{8}\right)^{\frac{1}{3}}$$

$$= 2 \left(\frac{y}{8}\right)^{\frac{1}{3}} \times \frac{1}{24} \left(\frac{y}{8}\right)^{\frac{2}{3}}$$

$$f(y) = \frac{1}{12} \left(\frac{y}{8}\right)$$

$$f(y) = \frac{8}{124} \quad 0 < y < 8$$

$$= 0 \quad 0 = w$$

Solution

$$3. f(x) = \frac{x}{12} \quad 1 < x < 5$$

$$= 0 \quad \text{o.w.}$$

pdf of $y = 2x - 3$

$$f(y) = f(x) \times \left| \frac{dx}{dy} \right|$$

$$\because y = 2x - 3 \Rightarrow x = \frac{y+3}{2}$$

$$\frac{dx}{dy} = \frac{1}{2}$$

Range for y

x	y
1	2
5	4

$$f(y) = f(x) \times \left| \frac{dx}{dy} \right|$$

$$= \frac{x}{12} \times \frac{1}{2}$$

$$f(y) = \frac{x}{24} \quad \because x = \frac{y+3}{2}$$

$$\therefore f(y) = \frac{1}{24} \left(\frac{y+3}{2} \right) = \frac{y+3}{48}$$

$$\therefore f(y) = \frac{y+3}{48} \quad 2 < y < 4$$

$$= 0 \quad \text{o.w.}$$